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Automated Metamorphic-Relation Generation with ChatGPT: An Experience Report

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***Abstract*—This paper reports on a pilot study of using ChatGPT, a language model based on GPT-3.5 architecture, for automatic generation of metamorphic relations (MRs), in the context of testing of autonomous driving systems (ADSs). The oracle problem is a major challenge in testing such systems, where it is difficult to determine whether or not the output of a system is correct. Metamorphic testing (MT) can alleviate this problem by checking the consistency of the system's outputs under various transformations. However, manual generation of MRs is often a time-consuming and error-prone process. Automated MR generation can yield several benefits, including enhanced efficiency, quality, coverage, scalability, and reusability in software testing, thereby facilitating a more comprehensive and effective testing process. In this paper, we investigate the effectiveness of using ChatGPT for automatic generation of MRs for ADSs. We provide a detailed methodology for generating MRs using ChatGPT and evaluate the generated MRs using our domain knowledge and existing MRs. The results of our study indicate that our proposed approach is effective at generating high-quality MRs, and can significantly reduce the manual effort required for MR generation. Furthermore, we discuss the practical implications and limitations of using ChatGPT for MR generation and provide recommendations for future research. Our study contributes to the advancement of automated testing of ADSs, which is crucial for ensuring their safety and reliability in real-world scenarios.**

***Index Terms*—Autonomous driving system (ADS); metamorphic testing (MT); metamorphic relation (MR); oracle problem; ChatGPT; natural language processing (NLP); large language model (LLM)**

I. Introduction

Autonomous driving systems (ADSs) have emerged as a promising technology for improving road safety and reducing traffic congestion. As ADSs become more sophisticated and complex, ensuring their correctness and reliability becomes a critical challenge [1]. Conventional testing methodologies such as unit testing and integration testing are often deemed insufficient in ensuring the safety and reliability of autonomous driving systems (ADSs), given their inability to capture critical safety scenarios. Furthermore, the testing of ADSs is vulnerable to the oracle problem, which constitutes a fundamental challenge to the efficacy of traditional testing approaches [2]. The oracle problem refers to the challenge of determining the expected outputs for a given set of inputs, which can be difficult or even impossible for complex systems [2].

Metamorphic testing (MT) [3], [4] is a testing technique that addresses the oracle problem by using metamorphic relations (MRs) to generate test cases. MRs are logical relations among related inputs and outputs that can be used to automatically generate test cases and increase the effectiveness of testing in various domains, including for ADSs [5]. For example, an MR for an ADS module that detects pedestrians might state that the module's output should be invariant under changes to the lighting conditions or the camera angles [6]. By applying MRs to the inputs and checking the outputs, we can detect potential faults in the system even when the oracle is unavailable.

However, MR specification can be a challenging and time-consuming task, especially for complex systems like ADSs. In traditional MT, MRs may be manually specified by human testers based on their domain knowledge and intuition. This process can be error-prone and may miss some important MRs due to human biases and limitations [3]. Moreover, manual MR specification may not be scalable for large and complex systems, which may have hundreds or thousands of MRs [4].

To address the challenge of manual MR specification, researchers have proposed automatic approaches for generating MRs [7], [8]. These approaches can reduce the effort and cost of MR specification and potentially improve their coverage and effectiveness. One approach is to use search-based techniques to explore the space of possible MRs and identify those that are most relevant and effective for the system under test (SUT) [9]. An interesting example of leveraging machine learning techniques in software testing is the development of a white-box method to predict the prevalence of predefined MRs for numerical programs [10]. This approach involves extracting the control flow graph of a function and training a predictive model using a code corpus of 48 mathematical functions. By doing so, the model can accurately predict the occurrence of permutative, additive, and inclusive MRs, showcasing its effectiveness in MR identification. This demonstrates the potential of utilizing machine learning for the identification and application of MRs in software testing, contributing to the development of efficient and reliable software [10].

In this paper, we report on a preliminary study exploring the use of ChatGPT, a state-of-the-art large language model [11],

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for automatic MR generation for ADSs. Specifically, we ask ChatGPT to generate MRs for several example ADS modules, and evaluate the effectiveness and efficiency of the generated MRs. ChatGPT is a powerful language model that has been trained on a large corpus of text and can generate human-like text based on a given prompt [12]. We hypothesize that ChatGPT can be used to generate high-quality and diverse MRs [13] for ADSs with faster speed than existing approaches [14], and that the generated MRs can improve the coverage and effectiveness of traditional manual MRs.

The rest of the paper is organized as follows. Section II provides background information on MT and automatic MR generation. Section III describes our approach for using ChatGPT to generate MRs for ADSs. Section IV presents an evaluation of the generated MRs, including comparisons with manually-specified MRs and an analysis of the difficulty of using ChatGPT. Section V discusses the limitations of our approach and future directions for research. Finally, Section VI concludes the paper.

II. Background

Autonomous Driving Systems (ADSs) are complex computer-controlled systems that aim to operate vehicles without human intervention through the use of various sensors, processors, and algorithms [1]. ADSs have the potential to significantly improve road safety and transform transportation. However, due to the intricate nature of their development and operation, extensive testing is required to ensure their reliability, safety, and effectiveness. Traditional testing methods may struggle to verify their correctness [15]. One of the main challenges in testing ADSs is the oracle problem, which refers to the difficulty of determining whether or not the system's outputs are correct [2]. This problem arises because it is often impractical to have a ground truth for the system's behaviour. In other words, we cannot always determine the correct output for a given input.

MT is a promising technique for testing ADSs as it allows for the detection of faults without relying on an oracle [3]. Instead, MT uses MRs, which are properties that should hold among related inputs and outputs of a system [4]. These relations are used to generate new test cases that explore the correctness of the system. For example, if we have an MR that states that the output of a system should remain the same after rotating the input 90 degrees, we can generate new test cases by changing the input and checking if the output remains the same.

Traditionally, MRs are manually crafted by domain experts, which can be a time-consuming and error-prone process [5]. This has led to a growing interest in the automatic generation of MRs. Researchers have proposed various approaches to automatically do this, including machine learning-based methods [10], semantic similarity-based methods [7], and constraint-solving-based methods [16]. While these approaches have shown promising results in some cases, there is still a need for more research to improve their effectiveness and scalability. The automatic generation of MRs represents a burgeoning and active research domain that holds significant promise for enhancing the efficiency and effectiveness of MT. However, to achieve robust and scalable results, further investigation and development are necessary.

ChatGPT is a language model based on the GPT-3.5 architecture and trained by OpenAI [17]. As a large language model, ChatGPT is capable of generating human-like responses to a wide range of natural language inputs, including questions, statements, and commands. It has been trained on a massive corpus of human text data from various sources, allowing it to understand and generate responses on a broad range of topics [11]. ChatGPT has been designed to perform a variety of language-related tasks, including natural language processing (NLP), text classification, and text generation. Its sophisticated algorithms and deep neural networks enable it to produce highly accurate and relevant responses to user queries, making it a valuable tool for a variety of applications, including customer service, chatbots, and language learning platforms [12].

In this paper, we investigate the use of ChatGPT for generating MRs for ADS modules. We evaluate the quality and effectiveness of the generated MRs and compare our approach with existing methods. Our investigation makes a novel contribution to the area of automatic MR generation, and highlights the potential for NLP techniques to augment the efficiency and effectiveness of software testing.

III. Experiment Design

A. Generating MRs using ChatGPT

We used ChatGPT to generate MRs for the parking module of an ADS [18]. We varied the number of MRs generated by ChatGPT and evaluated the quality of the generated MRs based on our own experiences using and understanding the parking module. Specifically, we assessed the usefulness and relevance of the MRs in testing the ADSs' parking module and their potential to reveal faults or weaknesses in the system.

B. Evaluating ChatGPT

To evaluate the effectiveness of the MRs produced by ChatGPT, we devised the marking scheme shown in Table I. This scheme includes a set of evaluation criteria, comprising: correctness, applicability, novelty, clarity, relevance to safety, overall usefulness, and computational feasibility. Specifically, correctness refers to the degree to which an MR is free of errors and accurately captures the intended behavior of the system. Applicability evaluates how widely an MR can be applied to different inputs and situations. Novelty assesses the originality and creativity of the MR. Clarity measures the ease of understanding and interpretation of the MR. Relevance to safety assesses how well the MR addresses safety-critical aspects of the system. Overall usefulness evaluates the practical utility of the MR for testing and verifying the system. Computational feasibility measures the ease and efficiency of applying the MR in practice.

Using these criteria, testers and researchers can gain insights into the strengths and weaknesses of the MRs generated by

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TABLE I: Marking table for evaluating the quality of metamorphic relations (MRs)

Criteria	Description	Score
Correctness	Does the MR accurately capture the intended behavior?	0-5
Applicability	Is the MR applicable to a wide range of inputs?	0-5
Novelty	Is the MR a new and original idea?	0-5
Clarity	Is the MR easy to understand and unambiguous?	0-5
Relevance to safety	Does the MR address a safety-critical aspect of the module?	0-5
Overall usefulness	How useful is the MR for testing the module?	0-5
Computational feasibility	Is it computationally feasible to apply the MR?	0-5

ChatGPT, and can make informed decisions about how to optimize their testing strategies.

IV. Experiment Results

A. Evaluating the quality of MRs generated by ChatGPT for small and large numbers of MRs

To assess the effectiveness of ChatGPT in generating MRs, we conducted a study wherein the model was requested to produce a diverse set of MRs, ranging from a small to a large number, for the SUT. The motivation for this exercise stems from the challenges faced by MT practitioners to formulate an adequate number of MRs. Our aim was to assess the diversity of MRs generated by the model to determine its capabilities.

In order to ensure that the comparisons were more direct and accurate, we made the requests to ChatGPT in separate conversational sessions. We also took into consideration the fact that ChatGPT is able to utilize context from previous conversations to inform its responses, thereby enabling it to generate more personalized and contextually-relevant outputs [17]. The outcomes produced by ChatGPT are presented in Table II. Due to the page limits, the table that contains a large number of MRs when ChatGPT has been asked can be found in the online repository[1].

It was observed that asking ChatGPT to generate a smaller number of MRs produces better and more accurate results than asking it to generate a larger number for the same ADS module. For example, the MRs in Table II contain the scenario description, and the relations among the inputs and outputs; while the results by requesting ChatGPT to generate a large number of MRs[1] are not correct, and lack such relations — they are more like function specifications.

We hypothesise that the reason behind this is that ChatGPT uses a probabilistic approach to generate responses [11], and when asked to generate a large number of MRs, there is a higher chance of some MRs being irrelevant or less useful for testing. However, when generating a smaller number of MRs, ChatGPT may have more resources to focus on each individual MR, leading to a higher likelihood of generating effective and relevant MRs. Therefore, when using ChatGPT to generate MRs for software testing, it is recommended to ask for a smaller number of MRs to improve the quality and accuracy of the generated MRs. While automated techniques for generating MRs offer various benefits, validating their correctness and relevance with domain experts before utilizing them for software testing is necessary. The lack of such validation increases the likelihood of generating irrelevant or erroneous MRs, which may result in the failure to identify defects, or the production of false positives: testers may spend significant time and effort resolving MR violations that are actually caused by incorrect MR definitions, leading to inefficiencies in the testing process. Hence, it is essential to integrate expert feedback and judgment into the MT process to ensure the efficacy and dependability of the generated MRs [4].

The quality of MRs generated by ChatGPT can vary depending on various factors, such as the complexity of the SUT and the specificity of the testing scenario. Generally, the efficacy of MRs generated by ChatGPT is lower when compared to existing MR-generation tools and strategies, such as Metamorphic Relation Input Patterns (MRIPs)[2] [19]; or even human-based approaches, which allow for customization based on different scenarios. This may be due to a lack of diversity and creativity in the generated MRs. One illustrative instance of human-generated MRs is the process of amalgamating two distinct scenarios wherein the SUT produces identical outputs. The resultant composite scenario ought to preserve the property of invariant behavior such that the SUT yields the same output.

While the MRs generated by ChatGPT may be useful for initial testing phases, due to their closeness to the functional requirements, their efficacy may be limited due to the small degree of change they introduce and the narrow range of input-output relationships explored [20]. Therefore, for complex systems requiring comprehensive testing and the generation of diverse and creative MRs, additional manual intervention may be necessary. This can enable exploration of the full spectrum of input-output relationships, thereby enhancing the overall effectiveness of testing [4].

B. Enhancing result accuracy through increased interaction with ChatGPT

Repeated conversations with ChatGPT have the potential to improve the accuracy of MRs generated. These interactions allow ChatGPT to learn from the past, and to refine its ability to generate effective and correct MRs [11]. As ChatGPT receives feedback about its answers, it can recognize patterns and relationships between inputs and outputs more accurately [20], leading to the generation of more comprehensive and diverse

[1] https://github.com/GabrielZZZ/2023STA_Appendix/blob/main/TableLongMRs.md

[2] A "Metamorphic Relation Input Pattern" is an abstraction that simplifies input generation for metamorphic testing by identifying common input characteristics among a set of metamorphic relations, which can be used to generate follow-up inputs for MT [19].

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TABLE II: The result of using ChatGPT for generating a small number of metamorphic relations

MRs	Description
MR1: Reverse into parking space	Given a scenario where the autonomous vehicle is in a parking lot and is required to park in a spot, the system should be able to reverse into the parking spot without hitting any other vehicle or obstacle in its path. The metamorphic relation is to test if the system is able to park in the same spot but with a different orientation, such as turning the vehicle 90 degrees or reversing into the spot from a different angle.
MR2: Parallel parking	Given a scenario where the autonomous vehicle is required to parallel park between two other parked cars, the system should be able to navigate into the parking spot and position itself correctly without hitting any other vehicle. The metamorphic relation is to test if the system is able to parallel park in a different spot, with different-sized vehicles on either side.
MR3: Obstacle detection	Given a scenario where the autonomous vehicle is required to park in a parking lot, the system should be able to detect obstacles such as pedestrians, other vehicles or shopping carts, and avoid hitting them. The metamorphic relation is to test if the system is able to detect and avoid different types of obstacles, such as bicycles or large delivery trucks.
MR4: Parking spot occupancy	Given a scenario where the autonomous vehicle is required to park in a parking lot, the system should be able to identify empty parking spots and park in them. The metamorphic relation is to test if the system is able to identify a parking spot as occupied, and adjust its path accordingly to find another spot.
MR5: Changing parking conditions	Given a scenario where the autonomous vehicle is required to park in a parking lot, the system should be able to adapt to changing parking conditions, such as an obstruction in the path, a spot becoming unavailable or a vehicle leaving its spot. The metamorphic relation is to test if the system is able to adjust its path and behavior in real-time based on changes in the parking environment.

TABLE III: Improved MRs from Table II after interacting with ChatGPT

MRs	Input Relation	Output Relation
MR1: Reverse into parking space	The autonomous vehicle is in a parking lot and required to park in a spot by reversing into it.	The output should be the same as the source input, but with the vehicle parked in a different orientation.
MR2: Parallel parking	The autonomous vehicle is in a scenario where it's required to parallel park between two other parked cars, with different-sized vehicles on either side of the parking space.	The output should be the same as the source input, but with the vehicle parked between different-sized vehicles.
MR3: Obstacle detection	The autonomous vehicle is in a parking lot and required to park in a spot, with obstacles such as pedestrians, other vehicles, or shopping carts, and different types of obstacles, such as bicycles or large delivery trucks.	The output should be the same as the source input, but with the vehicle avoiding different types of obstacles.
MR4: Parking spot occupancy	The autonomous vehicle is in a parking lot and required to park in an empty spot, but with the parking spot occupied by another vehicle.	The output should be the same as the source input, but with the vehicle finding another empty spot to park in.
MR5: Changing parking conditions	The autonomous vehicle is in a parking lot and required to park in a spot, with changing parking conditions such as an obstruction in the path, a spot becoming unavailable, or a vehicle leaving its spot, and different changing parking conditions, such as a different obstruction in the path, a different spot becoming unavailable, or a different vehicle leaving its spot.	The output should be the same as the source input, but with the vehicle adapting to different changing parking conditions.

MRs. The quality of feedback provided by the user can largely determine how many user prompts are needed, and the quality of the generated MRs. According to our empirical findings with ChatGPT, typical MR-enhancing prompts include: soliciting more elaborate descriptions of relationships; requiring the incorporation of input-output associations; and mandating the production of accurate and well-formed MR structures.

A detailed example of the interactions with ChatGPT regarding the evaluation of responses can be found in the online repository[3], while Table III shows the refined MRs, evolved from the originals in Table II.

In order to provide effective feedback, users should be familiar with the concepts of MT and MRs, although they do not need to be experts in the SUT domain. If the feedback is of high quality, ChatGPT can quickly learn from it, and generate high-quality MRs in fewer prompts. Conversely, if the feedback is of low quality or lacks specificity, ChatGPT may require more prompts to generate effective MRs.

[3]https://github.com/GabrielZZZ/2023STA_Appendix/blob/main/conversation-history.md

Engaging ChatGPT in diverse conversational contexts can expose it to a broader range of scenarios and inputs, thereby enhancing the generation of more effective MRs [20]. Through repeated interactions, ChatGPT can refine its understanding of the domain-specific concepts of the SUTs, which can lead to the generation of more accurate MRs. This is particularly beneficial when testing complex software systems.

It is noteworthy, however, that the number of feedback inputs required by the model is contingent on the preceding responses, and may vary depending on the SUT and user requirements [20]: The optimal amount of interaction required to improve the accuracy of the MRs generated by ChatGPT requires further investigation. Additionally, it is important to consider the impact of incorporating user feedback into the MR-generation process, particularly in situations where the user's level of expertise or the complexity of the SUT may affect the quality of the MRs generated.

Overall, the ability of ChatGPT to learn from past interactions and refine its answers to generate effective and correct MRs makes it a valuable tool in the field of software testing [20]. By providing clear communication and guidance

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to users on providing effective feedback, the quality and accuracy of MRs generated through repeated interactions with ChatGPT can be further improved, leading to more effective and thorough software testing.

C. Evaluating ChatGPT-generated MRs

The evaluation was conducted by an experienced MT practitioner, specialized in testing ADSs and generating MRs. The evaluation results are presented in Table IV, based on the assessment of the MRs from Table II. The analysis shows that all MRs exhibit high computational feasibility scores, indicating their potential for practical implementation. However, the novelty scores for these MRs are relatively low, suggesting that they may not significantly differ from existing MRs. Notably, MR3 received the highest total score (25) due to its high scores in correctness, applicability, relevance to safety, and computational feasibility, but scored low in novelty. MR5 received the second-highest total score (24) with high scores across all criteria except for novelty. MR1 and MR2 scored the lowest total scores (19 and 20, respectively) due to lower scores in several criteria. These results offer insights into the strengths and weaknesses of the evaluated MRs, highlighting potential areas for further improvement.

V. Discussion

A. Threats to Validity

The use of ChatGPT for generating MRs is a promising approach, but its limitations and potential threats to validity must be considered. Customization and refinement may be necessary to ensure that the generated MRs are relevant and appropriate for the specific SUT under consideration. In addition, the diversity of the MRs generated may be limited, which could reduce the effectiveness of the testing approach, particularly when generating a large number of MRs.

User feedback used to refine MRs may be subjective, and individual biases or errors may negatively impact the quality of the generated MRs. The effectiveness of the testing approach may also vary depending on the complexity and variability of the SUTs, which reduces the generalizability of findings to other domains and SUTs. Moreover, the reproducibility of the approach is a potential limitation as it assumes a high level of expertise on the part of the person interacting with ChatGPT, which may reduce the replicability of the results. Additionally, the refinement process through continuous conversation may limit the creativity that ChatGPT could offer as it adjusts its responses to the inquirer. Furthermore, the reliability of the feedback mechanism may be burdened with psychological limitations, as different people may focus on different aspects of the phenomenon under study.

B. Future Work

Although our study has demonstrated the potential of ChatGPT for the automatic generation of MRs for testing ADS modules, further research is necessary to advance this technique and maximize its benefits for software testing. The following are some potential future research directions:

1) Enhancing the quality analysis of ChatGPT-generated MRs by involving more evaluators and refining the marking criteria. This could include conducting more comprehensive empirical studies and user surveys to obtain a more diverse range of feedback and perspectives on the quality of the generated MRs.
2) Developing more systematic guidelines for interacting with ChatGPT to improve the quality of MRs. This could involve providing training and support for testers on how to refine and customize the generated MRs for specific SUTs, as well as developing best practices and guidelines for evaluating and selecting the most appropriate MRs.
3) Investigating various methods for improving the diversity of ChatGPT-generated MRs, such as utilizing different input prompts, generating more variations of responses, and integrating external knowledge sources. This could include exploring different techniques for fine-tuning the model and pre-processing the input data to enhance the diversity and quality of the generated MRs.
4) Developing tools and frameworks to facilitate the customization and refinement of ChatGPT-generated MRs for specific SUTs. This could include designing user-friendly interfaces and visualization tools to enable testers to interact with the generated MRs and provide feedback for further refinement.

VI. Conclusion

In conclusion, this preliminary study has demonstrated the potential for using ChatGPT to generate MRs for testing ADSs. The results suggest that ChatGPT can be a cost-effective and time-saving approach for generating diverse and relevant MRs. However, it is important to note that the quality and relevance of the generated relations may vary, and additional customization and refinement by the testing team may be necessary. Overall, ChatGPT shows promise as a tool for MR generation in software testing, and further research is needed to explore its full potential in this area.

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TABLE IV: Evaluation results of MRs from Table III

Marking Criteria	**MR1**	**MR2**	**MR3**	**MR4**	**MR5**
Correctness	2	3	4	3	4
Applicability	4	4	4	4	4
Novelty	1	1	1	1	1
Clarity	2	2	4	3	3
Relevance to safety	2	2	4	4	4
Overall usefulness	3	3	3	3	3
Computational feasibility	5	5	5	5	5
Total score	19	20	25	23	24
Explanation	The MR has low correctness and clarity scores due to its dependence on functional specifications affected by vehicle orientation. However, it has high applicability and computational feasibility. Its usefulness score is moderate, and novelty score is low, resembling functional requirements.	The MR scores moderately in all aspects, except for computational feasibility, which is high. However, its correctness is partially affected by the lack of functional requirements determining obstacle size. While it is generally applicable, clear, and relevant to safety, it lacks novelty and usefulness.	The MR shows high correctness, applicability, relevance to safety, and computational feasibility. Its clear explanation is a plus, but the low novelty score indicates similarity to functional requirements. The MR's moderate usefulness score implies potential value, but other MRs may be more impactful.	The MR shows moderate scores in all criteria, except for high applicability and relevance to safety. However, the SUT's behavior may vary depending on functional specifications like changes in the parking slot or aborting the function. The explanation is clear, and the implementation is computationally feasible. Yet, it lacks novelty and overall usefulness.	The MR scores high for correctness, applicability, relevance to safety, and computational feasibility. However, the low clarity score lacks sufficient details on how the vehicle adapts to changing parking conditions. The low novelty score aligns with the functional requirements of the SUT. Overall usefulness is moderate, requiring further development.

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